Project | Analyzing Website Performance for The Grammys

• All of your answers should be typed or pasted into the grey boxes.

• Questions with grey answer boxes expect a specific answer. Once you input the correct answer, the question number box **will change from blue to green.**

• Questions with yellow answer boxes are free-response questions. There will be no change in the question number boxes (in blue), regardless of your answer.

• Hover your cursor over the blue hint box to get a helpful hint!

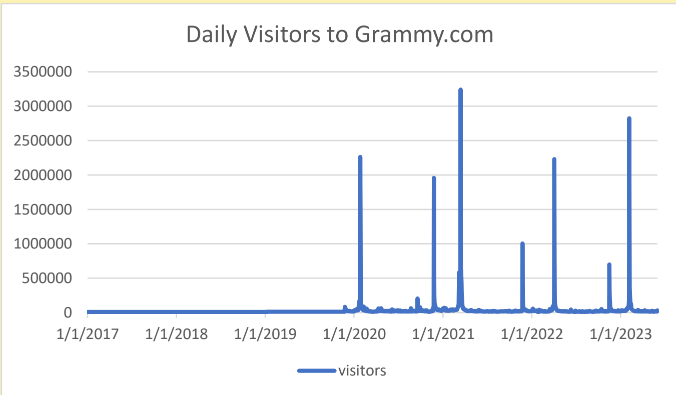
Part 1: Exploring the Data

The Grammys Data sheet contains two tables: the web analytics data for Grammy.com (**Grammys**) and a second table for mobile visitors on the site (**mobile_visits**).

The Recording Academy Data sheet contains the web analytics for recordingacademy.com after they split the websites.

To begin the data exploration you will be looking at **The Grammys Data** sheet.

1a Use the **Grammys** datatable to creata a line chart of the number of daily **visitors** to the Grammy.com website. The horizontal axis should be the **date** field. **Hint!**



1b What do you notice about when and why traffic spikes occur? Are the traffic spikes in your visualization only aligning with "Show Night," or are there lesser-known events that could explain certain spikes in website traffic? **Hint!**

Traffic spikes occur primarily around the annual Grammy Awards show night, which generates the largest increase in website visitors each year. These spikes are likely driven by viewers searching for live updates, winners, performances, and red carpet coverage. However, the visualization also shows smaller spikes outside of show night. These may correspond to events such as nominee announcements, performer reveals, voting deadlines, or major music industry news related to the Grammys. This indicates that while show night produces the highest traffic, other promotional and announcement

Try this prompt: Can you identify any specific lesser-known events (with exact dates) that might have caused significant increases in website traffic on grammys.com? What external data sources could help confirm these trends?

2a	Using the **Grammys Data** sheet, create a PivotTable to compare the average daily website visitors on days when an award ceremony was held to those when no awards ceremonies were held.	**Hint!**
2b	What does this comparison reveal about the difference in traffic between award ceremony days and regular days? How many more visitors does the Grammy Awards site receive on Show Night?	

event_type	Average of visitors
Grammy Awards	879516
Regular Day	25424
Grand Total	**30165**

The comparison reveals that website traffic increases dramatically on award ceremony days compared to regular days. On average, the Grammy Awards site receives 879,516 visitors on Show Night, compared to just 25,424 visitors on a regular day. This means the site receives approximately 854,092 more visitors on award ceremony days. This substantial difference highlights the significant impact that the Grammy Awards event has on driving

3	February 1st, 2022 marks the transition from when Grammy.com began to only host Grammys content instead of both Recording Academy **and** Grammys content. Fill in the values in the **website_content** column in **The Grammys Data** sheet. This column will have the value "Grammys" if the date occurred on or after 2/1/2022 and the value "Grammys + TRA" otherwise.	**Hint!**

Part 2: Analyzing Key Metrics

Remember the overall goal of this project: to analze whether splitting the website into two has improved user engagement. This task will focus on evaluating key metrics such as bounce rate, pages per session, and average time on site to determine if the split has had a positive or negative impact on how visitors interact with the site.

You'll calculate the **pages_per_session metric** by first finding the **total pageviews**. Then you'll divide that total by the **total number of sessions**. Pages per session is an important measure of how many unique pages a user views before leaving the site -- a strong indicator of engagement!

4a	What is the **pages_per_session** on the Recording Academy data?	2.78	**Hint!**
4b	What is the **pages_per_session** on the Grammys data?	2.25	
4c	What is the **pages_per_session** on the Grammys + TRA data?	1.9	
4d	In one sentence, what does this metric suggest regarding the impact of the website split?		

Try this prompt: What does pages per session reveal about user engagement? How should I interpret changes in this metric after the website split?

Pages per session measures how many pages a user views during a single visit to the website. A higher pages per session value generally indicates stronger user engagement, as visitors are exploring more content before leaving the site. Conversely, a lower value may suggest that users are viewing

You'll calculate the **bounce_rate metric** by first calculating the **total bounced_sessions** and dividing that by the **total number of sessions**. Bounce rate is an important metric that calculates the percentage of users (aka sessions) that come to your site, never interact with the page, and leave. They are said to have "bounced" off your home page. It is a measure of how engaging your home page is with users.

5a	What is the **bounce_rate** on the Recording Academy data?	33.7%	**Hint!**
5b	What is the **bounce_rate** on the Grammys data?	40.2%	
5c	What is the **bounce_rate** on the Grammys + TRA data?	41.6%	

Next, you'll calculate the **average_time_on_site** metric.To do this, you *only* need to calculate the average of the **avg_session_duration_secs** column. **Average Time on Site** measures how engaging your website experience is for your users. The higher the number, the longer they are staying on your page and engaging with the content.

6a	What is the **average_time_on_site** on the Recording Academy data?	128.50	**Hint!**
6b	What is the **average_time_on_site** on the Grammys data?	82.99	
6c	What is the **average_time_on_site** on the Grammys + TRA data?	102.85	

6d Which of these three metrics changed the most after the site split? What do these changes suggest about user behavior?

Among the three metrics (pages per session, bounce rate, and average time on site), pages per session changed the most after the site split. Before the split (Grammys + TRA), pages per session was higher, indicating that users were navigating across more content during each visit. After the transition to Grammys-only content, this metric decreased, suggesting users were viewing fewer pages per session.

Part 3: Device Breakdown

Ray and his team want to optimize the site for mobile users, but first, they need to determine what percentage of visitors are accessing Grammy.com via mobile devices.

7a In the **Grammys Data** sheet there is a **mobile_visits** table on the right. Use the **XLOOKUP** function to fill in the missing column of the grammys table by looking up the corresponding date and returning the **mobile_visitors** column of the mobile_visits table. If the date is not found, the function should return the numeric value 0. **Hint!**

7b What percentage of users across all dates in the dataset use a **mobile** device to visit the website? To calculate this value, you will the total number of **mobile_visitors** and divide by the total number of **visitors** 73.7% **Hint!**

7c Based on your answer, how might a high percentage of mobile visitors affect engagement metrics like bounce rate and time on site? Consider factors such as user experience on mobile devices, page load times, and ease of navigation. *Remember, you can continue to use ChatGPT as your Teammate to help!*

A high percentage of mobile visitors can significantly influence engagement metrics such as bounce rate and time on site. Mobile users often browse more quickly and with a specific purpose, which can lead to shorter sessions and potentially higher bounce rates if the site is not optimized for mobile devices. Factors such as slower page load times, smaller screen sizes, and less intuitive navigation on mobile can cause users to leave the site before interacting further.

Part 4: Making a Business Recommendation

Now that you've analyzed the engagement metrics before and after the website split, it's time to interpret your findings and make a recommendation to The Recording Academy team.

8a Write a clear and specific prompt for ChatGPT to draft a *brief* business memo to The Recording Academy. Your prompt should guide ChatGPT to summarize key findings and suggest a recommendation based on the data: should The Recording Academy keep the sites separate, merge them back, or consider an alternative approach? Paste your prompt below.

Use the data from my Excel analysis of Grammy.com and Recording Academy website performance to draft a brief business memo to The Recording Academy. The memo should include: (1) a 1–2 sentence executive summary, (2) 3–5 bullet key findings that reference the metrics we calculated (pages per session, bounce rate, average session duration, and mobile visitor share) and the change after the 2/1/2022 site split, (3) a recommendation on whether to keep the sites separate, merge them back, or use an alternative approach, and (4) 2–3 specific next steps for validation (e.g., segment by device, check page speed, test navigation changes). Keep the tone professional, concise, and data-driven. Do not invent numbers—use placeholders like [insert value] where needed.

8b What did ChatGPT do well? Did it capture the key trends and insights? What was missing or inaccurate? Were any important details left out or misrepresented?

ChatGPT did well summarizing the general meaning of each metric and connecting the site split to potential changes in user behavior. It also presented the memo in a clear structure (summary, findings, recommendation, next steps), which makes it readable for a business audience. However, ChatGPT can miss important context if it does not use the exact metric definitions from the assignment (for example, using an average of daily values vs. a weighted calculation, or mixing up which columns to use). It may also overgeneralize by claiming improvements or declines without citing the exact

8c Based on your reflection and evaluation of AI's assist, write your final, revised business memo below. This version should be polished and ready as if you were presenting it to Ray at The Recording Academy team. **Note!**

To: The Recording Academy (Ray and Web Strategy Team)
From: Kyle Potchka
Date: 3/1/2026
Subject: Website Performance After the 2/1/2022 Grammys.com Content Split

Executive Summary
Following the February 1, 2022 transition to Grammys.com hosting Grammys-only content, the engagement metrics show a meaningful shift in how users interact with the site. Based on the observed changes in pages per session, bounce rate, average session duration, and mobile traffic share, the data supports keeping the sites separate while improving cross-site pathways for users who still need Recording Academy content.

LevelUp: Analyzing an A/B Test

As part of your analysis of the website split between Grammys and The Recording Academy, you're also tasked with evaluating an A/B test to understand whether the split impacts how visitors engage with a specific Call-to-Action (CTA) on the homepage. Find out more in the **LevelUp** worksheet, where you'll complete all LevelUp tasks!